UE23CS441A- Capstone Project Phase - 3

Project Progress Review 3

Project Title: BioPathNet++ for Autoimmune Diseases: Path-Based Transfer Learning for Rare Disease Drug Repurposing

Project ID: PW26_PH_03

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Abstract

Problem. Rare autoimmune diseases (PBC, systemic sclerosis, Sjögren, autoimmune hepatitis, myasthenia gravis, Crohn) have few approved drugs. Repurposing must work **zero-shot**: rank existing drugs for a disease whose treatments the model has never seen.

Approach. BioPathNet++: a path-based graph neural network (Neural Bellman-Ford) on PrimeKG (129,375 nodes, 4 million edges) plus a leakage-free transfer channel, trained on free Kaggle T4 GPUs.

Rigour. An audit showed our earlier near-perfect scores came from leakage. We rebuilt everything: strict-family hold-out, full ranking of all 7,957 drugs, six baselines, 3 seeds, paired tests, TxGNN head-to-head, five validation layers.

Scope. Ranked hypotheses for expert review, not treatment advice.

76%

unseen test diseases with an approved drug in the top 10

best baseline 57%, random 0.8%

0.501 vs 0.392

MRR vs best baseline (68 test diseases)

p = 0.003, 3 seeds

7,957

drugs ranked for every disease

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Suggestions from Review - 2

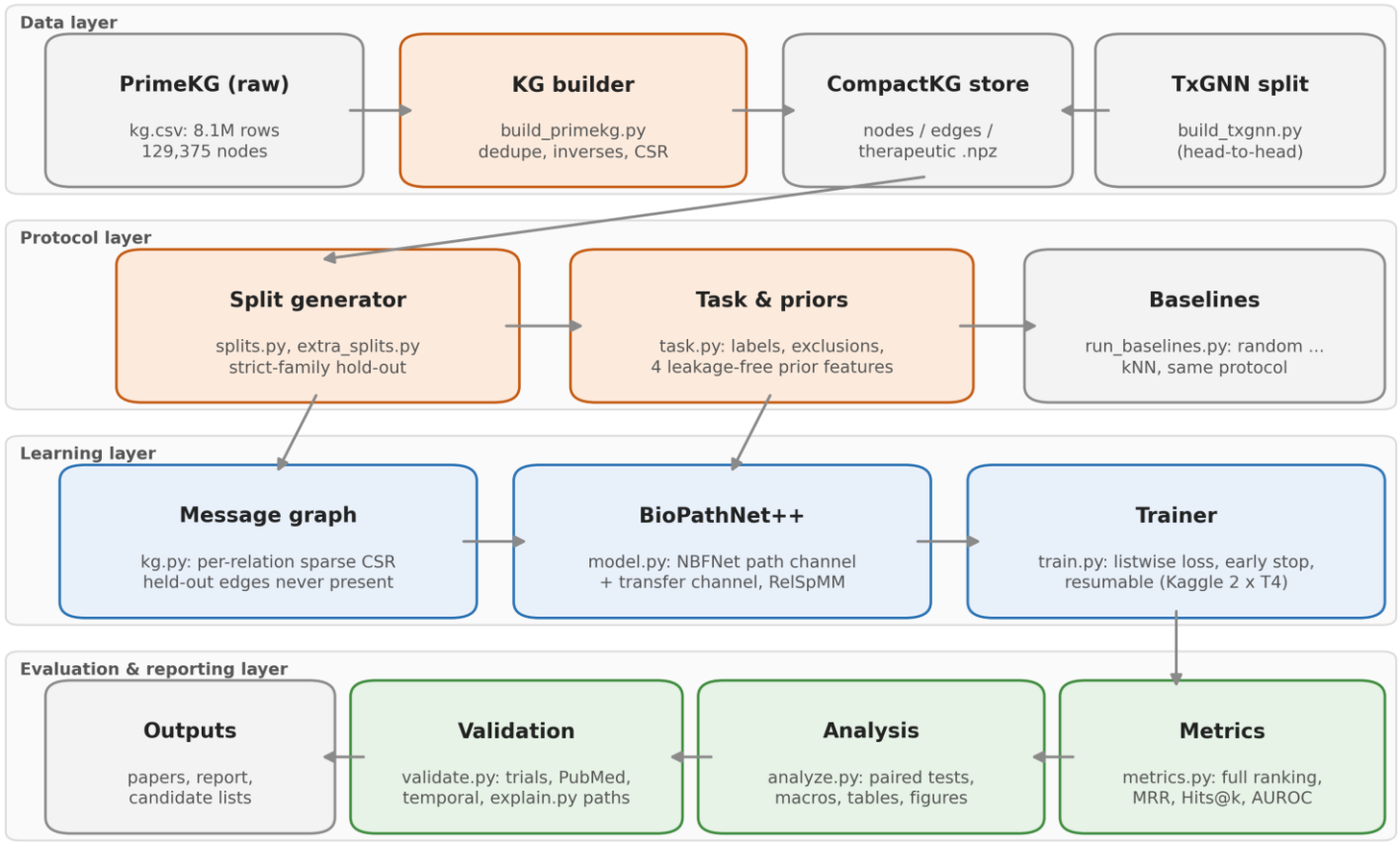
Panel remarks (Review 2)

[Add the panel's remarks from Review 2 and our response to each.]

Key changes made in Phase 3

- Audited the earlier pipeline: its AUPRC 0.935-0.999 came from leakage (277/277 test pairs were graph edges; a no-learning degree rule scored AUROC 0.972). Those numbers are withdrawn.
- Rebuilt on PrimeKG with a strict-family zero-shot protocol and full ranking.
- Added six baselines, 3 seeds, paired tests and a selection-free test split.
- Head-to-head with TxGNN on TxGNN's own split; five validation layers.
- IEEE conference and journal drafts; college Turnitin similarity 0%.

Implementation Details: System Architecture

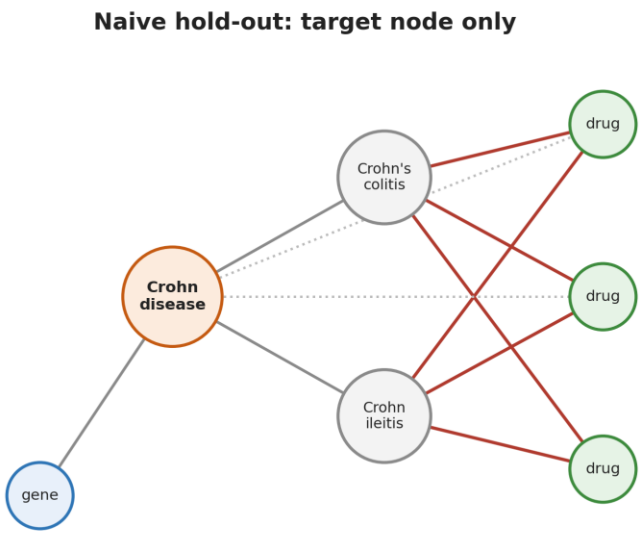


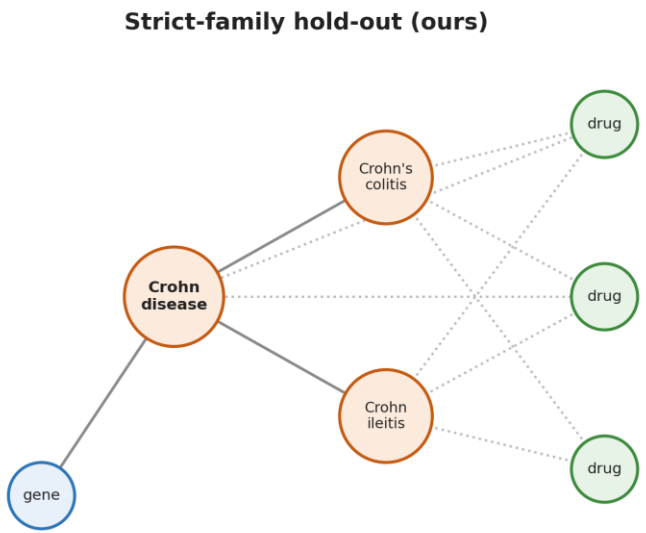
Technology

- Python 3.13, PyTorch sparse CSR
- PrimeKG knowledge graph (Harvard)
- Kaggle notebooks, 2 x NVIDIA T4
- pytest, Git / GitHub
- DGL 2.4 in a separate env (TxGNN)
- ClinicalTrials.gov and PubMed APIs

Package **bpp**: 19 modules, about 3,900 lines; every paper number is generated from run logs.

Implementation Details: Leakage-Free Protocol





A disease is held out together with its whole family (disease-disease neighbours). Without this, all 16 Crohn drugs stayed reachable via "Crohn's colitis".

Why: audit of the old pipeline

277 / 277

test pairs were edges of the graph

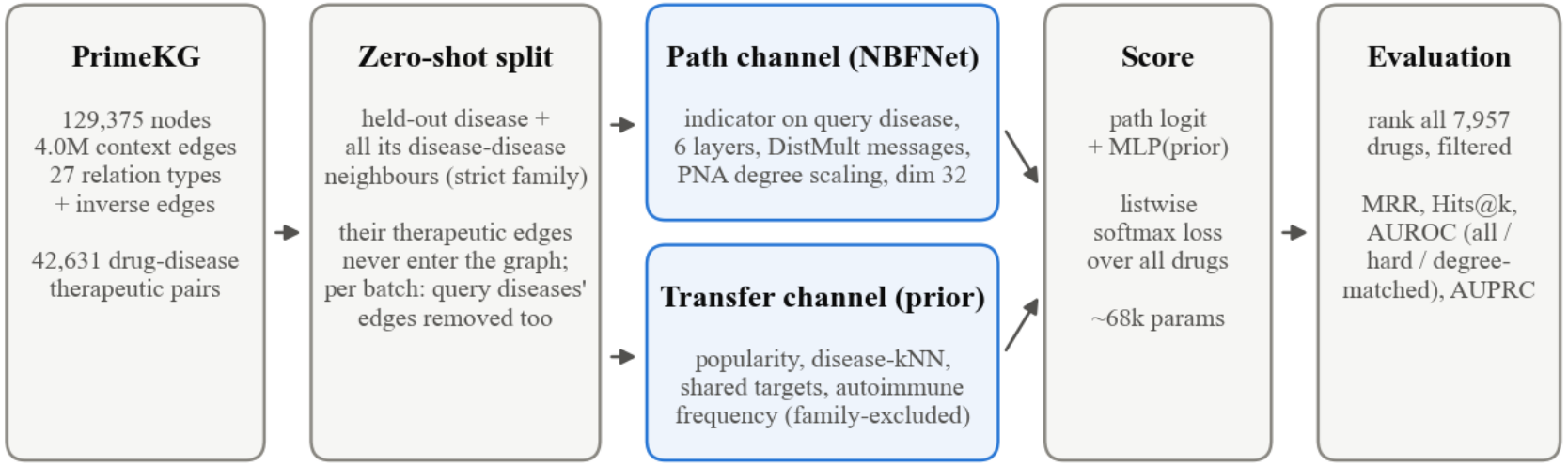
0.972

test AUROC of a no-learning degree rule (model 0.941)

5 / 6

targets unreachable: same drug ranked first everywhere

Implementation Details: Model



both channels see only training diseases' drug labels

Path channel (NBFNet)

- Starts at the query disease, 6 layers
- Relation-specific DistMult messages
- Scores all 7,957 drugs in one pass
- Needs no embedding of the disease

Memory trick (RelSpMM)

- 60 sparse relation matrices
- Recompute A_r H in backward pass
- Memory O(V x B x d), not O(E x B x d)
- Full 4M-edge graph: peak 5.8 GB on a T4

Transfer channel + loss

- 4 leakage-free priors (popularity, kNN,
- shared targets, autoimmune frequency)
- Listwise softmax over all drugs
- 68,226 parameters in total

Implementation Details: Algorithm and Modules

Training loop (pseudocode)

```
for epoch in 1..20:
  for batch U of 8 training diseases:
    G_b <- graph without drug edges of families(U)
    h <- indicator on each u in U
    for t in 1..6:  h <- NBFLayer_t(h, G_b)
    s <- MLP(h at all drugs) + g(prior(U))
    loss <- listwise softmax(s, known drugs)
    backward; clip; Adam step
  validation MRR -> best.pt; last.pt
  stop after 6 epochs without gain
```

Module	Role
build_primekg.py	graph + splits (35 s)
splits.py	strict-family hold-out
task.py	labels, priors, baselines
model.py	RelSpMM, NBF layers
train.py	training, resume, early stop
metrics.py	full ranking, bootstrap
analyze.py	paired tests, paper numbers
validate.py	trials, PubMed, temporal
explain.py	path explanations
build_txgnn.py	TxGNN split converter

Project Demonstration

Example output: Myasthenia gravis (never seen in training)

Rank	Drug	Evidence
1	Ambenonium	★ known (PrimeKG)
2	Neostigmine	★ known
3	Pyridostigmine	★ known, phase 3/4 trials
4	Distigmine	★ known
5	Prednisolone	★ known
7	**Amifampridine**	**new: phase-3 trial NCT03304054**

★ = drug PrimeKG lists for MG; ranks averaged over 3 seeds. The model saw none of MG's drug links.

Why pyridostigmine? (path explanation)

myasthenia gravis → MUSK gene → neuromuscular junction disease → pyridostigmine

Gradient x input on every edge, then beam search for the strongest paths.

Live demo (laptop GPU)

```
python -m bpp.score --run <run> --out scores.npz
python -m bpp.explain --run <run> --out paths.json
```

Scoring reproduces the Kaggle top-10 lists on the laptop (17 of 18; one exact tie).

Test Plan and Strategy

Test activity	What it checks
Unit tests (pytest, 9)	gradcheck, edge removal, reachability
Leakage assertions	no test pair reaches training
Reproducibility checks	Kaggle code = repo; resumed runs
Statistical testing	3 seeds, bootstrap CI + Wilcoxon
Selection-free test split	68 diseases never used to tune
Local rehearsal	TxGNN pipeline on mini PrimeKG
External validation	trials, PubMed, later approvals

Why this approach

A research model fails silently: leakage gives great numbers, so tests target leakage and statistics.

Environment

Laptop (RTX 3050) for build, tests, analysis; Kaggle 2 x T4 for training.

Team roles

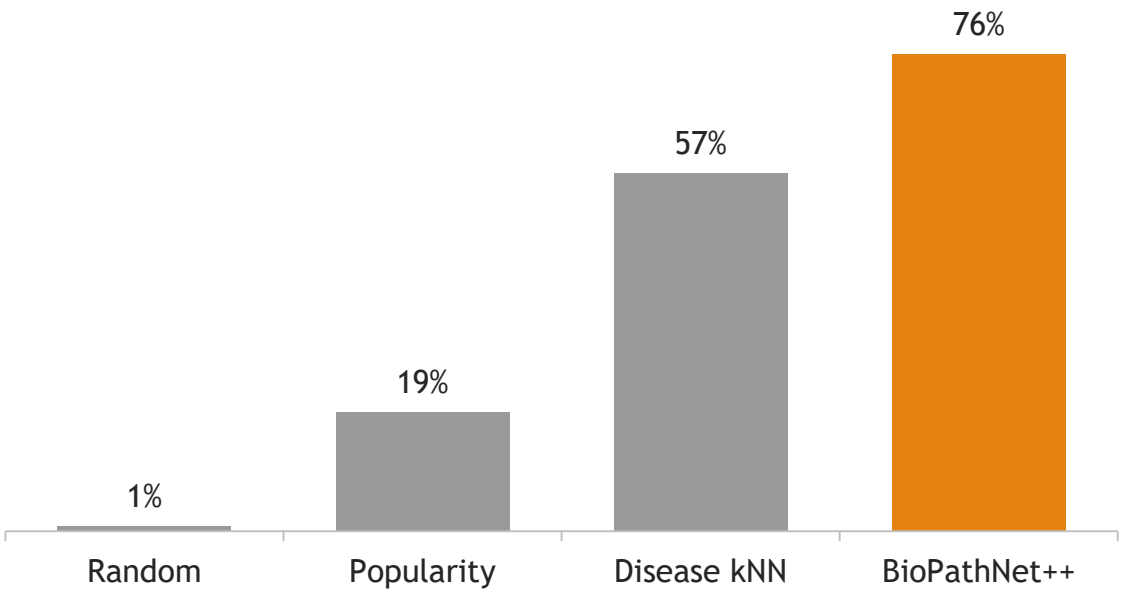
[Fill in each member's testing role]

Drawback

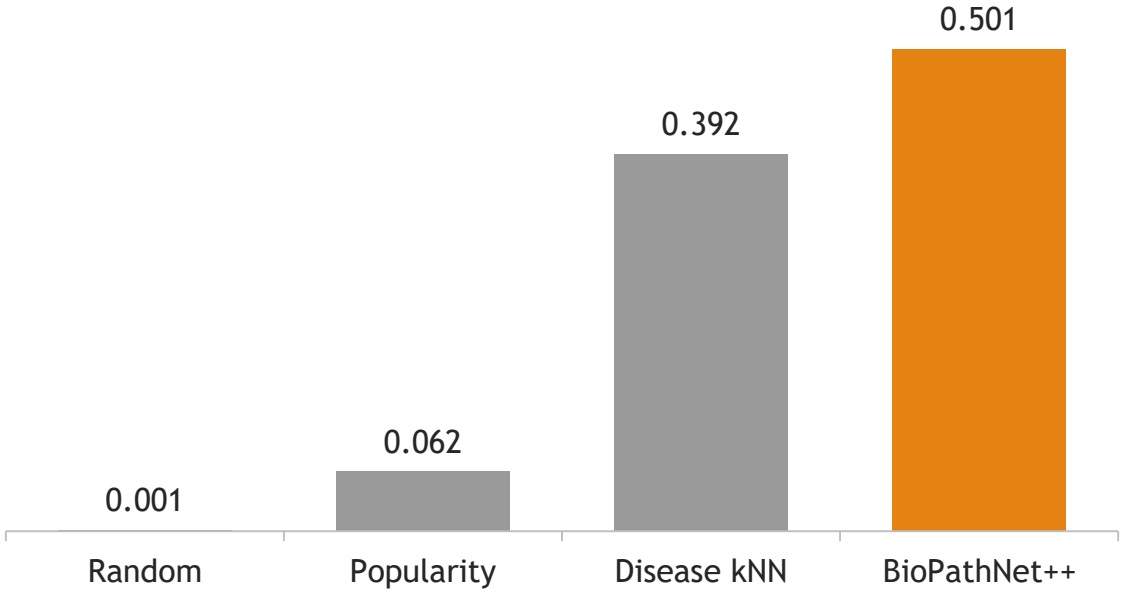
Small rare-disease sets give low statistical power.

Results and Discussion: Accuracy on Unseen Diseases

Approved drug in top 10 (% of 68 test diseases)

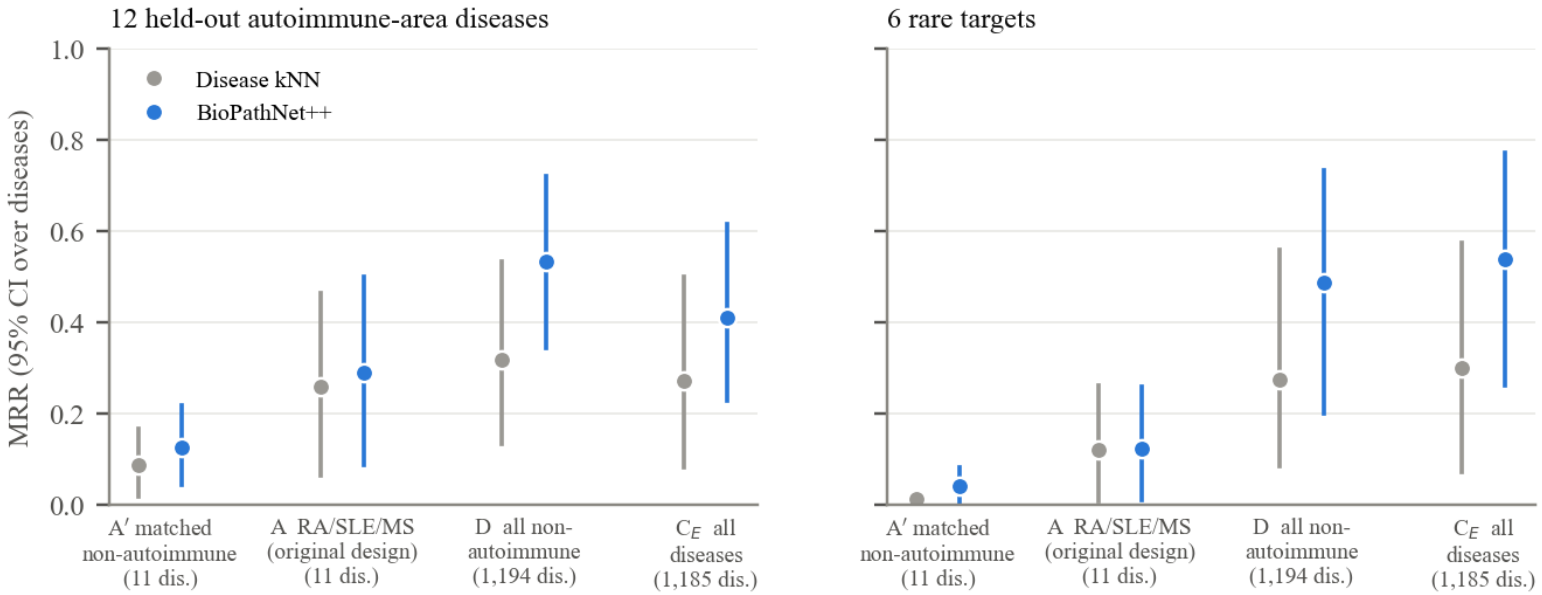


Mean reciprocal rank (68 test diseases)



Gain over the best baseline: **+0.109 MRR** (95% CI +0.044 to +0.180, p = 0.003, 3 seeds). Validation diseases agree: 0.394 vs 0.281 (p < 0.001). Median rank of a known drug: 2 of 7,957.

Results and Discussion: Source of Transfer (Aim 1)



MRR on 12 held-out autoimmune diseases (left) and the 6 rare targets (right) by training data. A' = 11 non-autoimmune diseases matched on number of labels.

Scarce data

RA/SLE/MS: **0.290** vs matched controls 0.126 (p = 0.016).

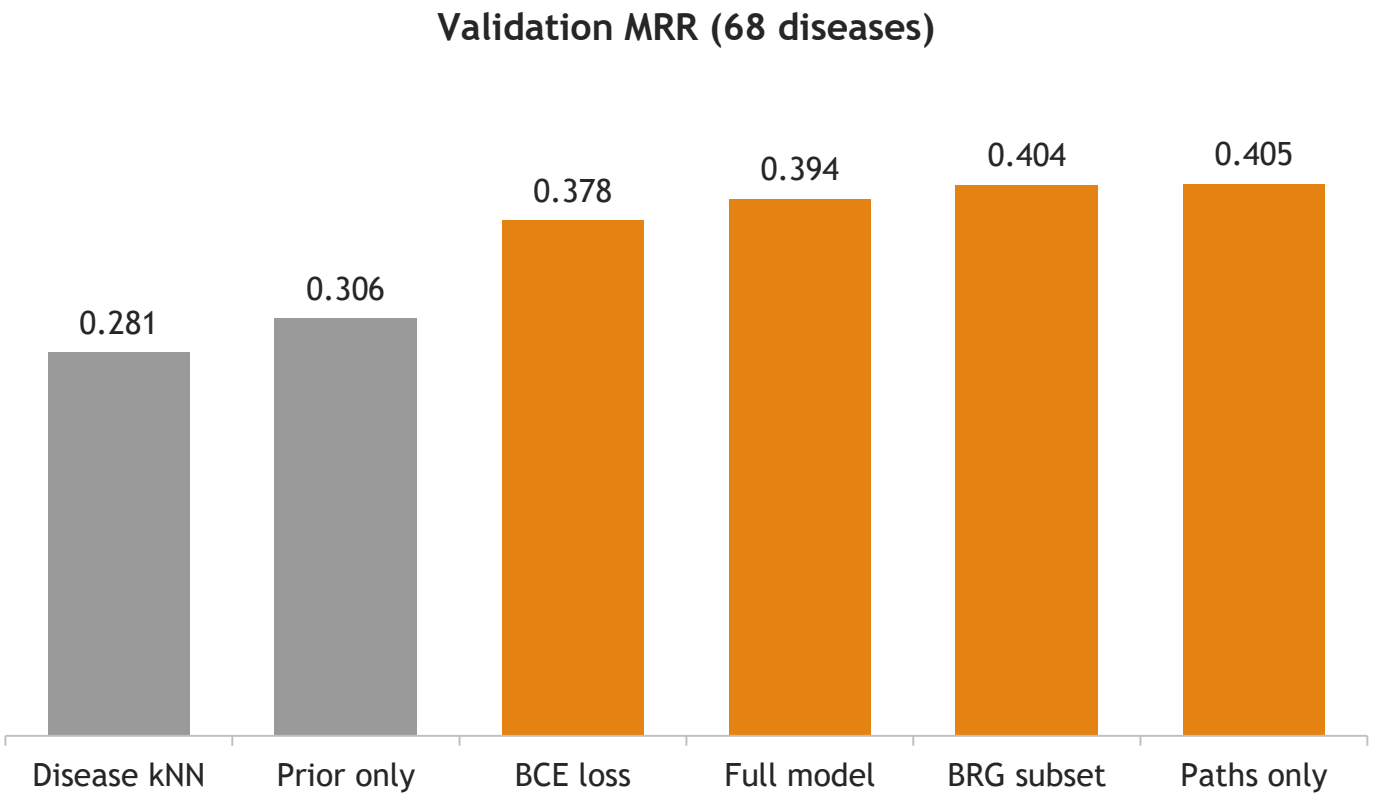
But paths alone: 0.007 vs 0.010, so the gain is a drug-class prior.

Full scale

Adding autoimmune data **lowers** MRR: 0.410 vs 0.532 (p = 0.010, 3 seeds).

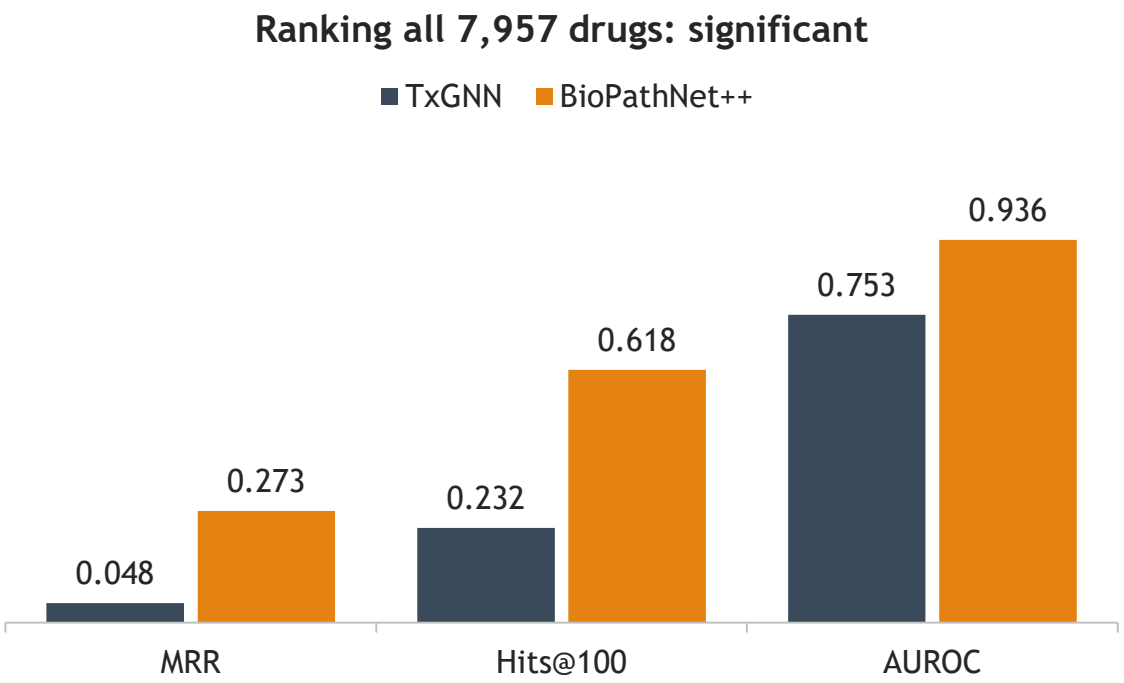
Rare targets unchanged without them (0.487 vs 0.475).

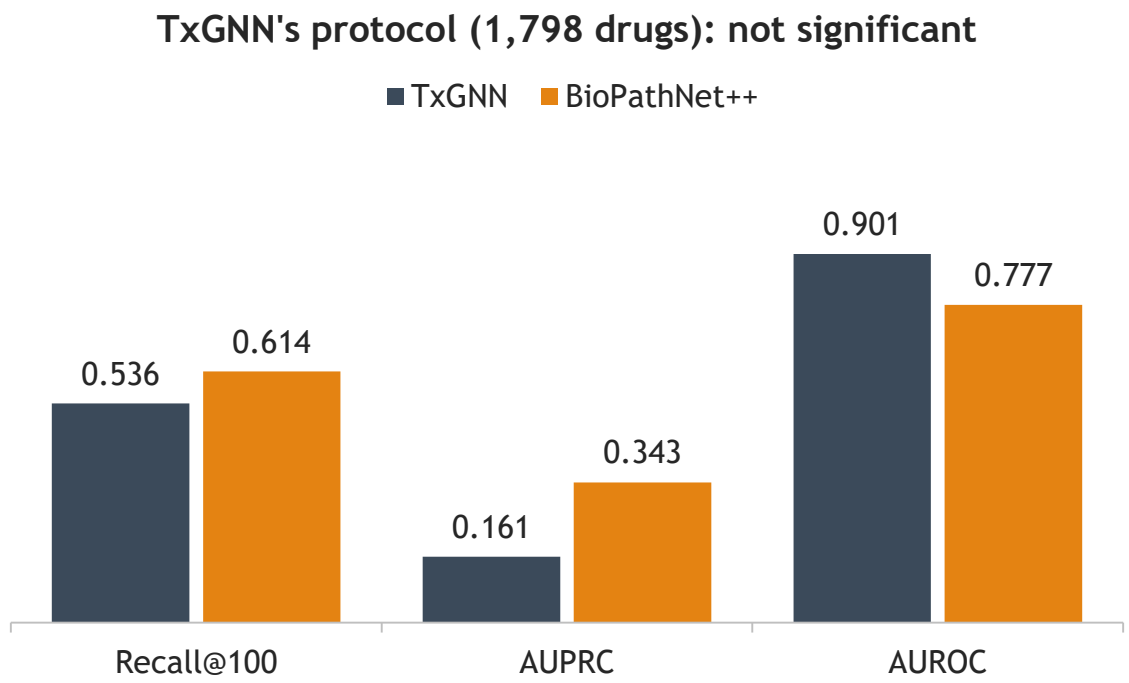
Results and Discussion: Component Analysis (Aim 3)



- **Path reasoning carries the accuracy**: without it MRR drops to 0.306 ($p < 0.001$).
- **Transfer channel = stability**: mean unchanged ($p = 0.15$), target sd 0.087 → 0.017 across seeds.
- **Loss choice is a tie**: BCE 0.378 vs listwise 0.394 ($p = 0.34$).
- **72% of edges are redundant**: 4.0M → 1.1M edges, same MRR, 11% faster.

Results and Discussion: Head-to-Head with TxGNN





TxGNN's own split and code, 13 test diseases, 3 seeds each; our model trained only on edges TxGNN also trains on. Why the protocols disagree: only **23%** of TxGNN's top-100 drugs have any indication (ours **99%**); TxGNN's protocol filters those drugs out.

Results and Discussion: Case Studies (Top 5 per Target)

Rank	PBC	SSc	Sjögren	AIH	MG	Crohn
1	Ursodeoxycholic acid★	Benzoic acid★	Propylene glycol★	Prednisolone	Ambenonium★	Hydrocortisone★
2	Cholic acid	Aminobenzoic acid★	Lifitegrast	Triamcinolone	Neostigmine★	Cortisone acetate★
3	Doripenem	Prednisolone [B]	Povidone★	Hydrocortisone acetate	Pyridostigmine★	Triamcinolone★
4	Homatropine	Triamcinolone	Povidone K30★	Sulfacetamide	Distigmine★	Prednisolone★
5	Ampicillin	Betamethasone	Prednisolone★	Hydrocortisone	Prednisolone★	Ofloxacin

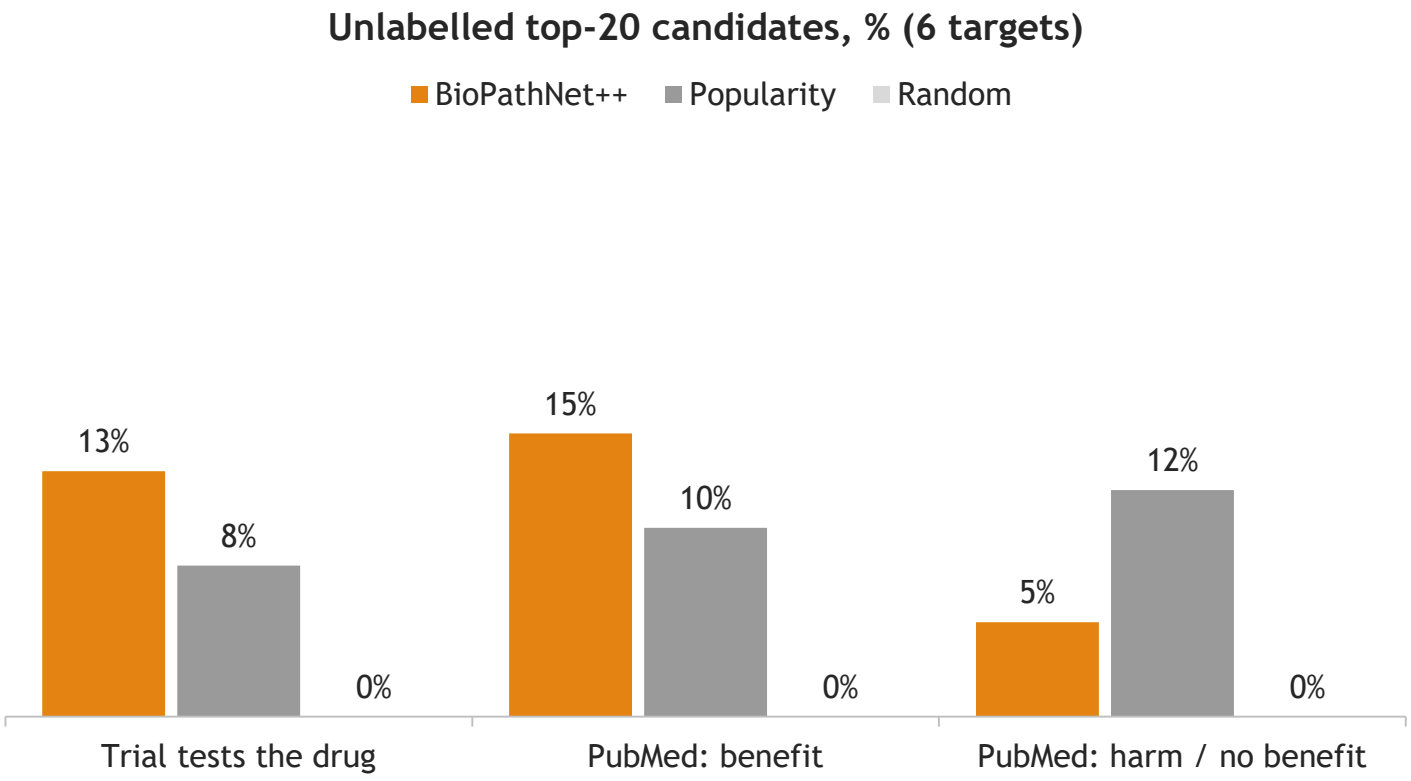
★ known in PrimeKG; [A]/[B] phase 3-4 / phase 2 trial that tests the drug in the disease; rank averaged over 3 seeds

PBC
First-line drug ursodeoxycholic acid ranked #1 in every seed.

MG
4 cholinesterase inhibitors at #1-4; amifampridine (#7) has a phase-3 MG trial.

AIH
Label gap: model's #1 prednisolone is standard therapy, but PrimeKG lists one other drug.

Results and Discussion: Validation of New Candidates



Temporal check

6 drugs approved after the data snapshot rank at a median of **1,104 / 7,957** (top 14%); risankizumab for Crohn ranks **#37**. Not significant (p = 0.11).

Honest reading

Candidates are plausible (13% have a trial testing the drug vs 0% for random drugs) but only modestly ahead of popular drugs. Clinician review is the next step.

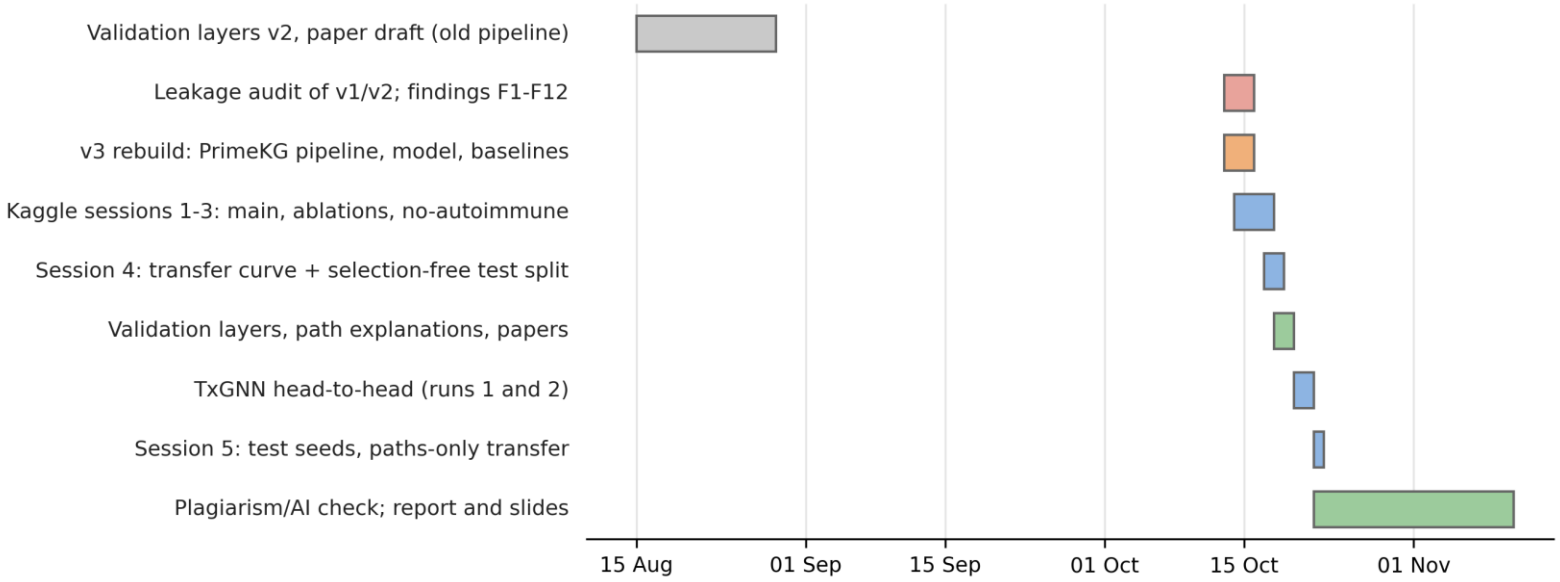
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Applicability to Other Domains

Domain	Common problem	What changes	Data source
Other rare disease areas	few known drugs per disease	only the held-out set	PrimeKG
Drug side effects	predict unseen drug-effect links	query relation	SIDER / PrimeKG
Drug-target discovery	new targets for a drug	source = drug node	DrugBank / PrimeKG
Clinical decision support	rank options with reasons	add path explanations UI	hospital KG
Auditing other KG models	leakage in evaluation	reuse protocol + baselines	any KG

Benefits: inductive (works for unseen nodes), explainable, fits on free GPUs. **Feasibility:** only the query relation and split change; code is relation-agnostic. **Limits:** needs a curated knowledge graph; incomplete labels lower measured accuracy.

Timeline: Update on Tasks



Status

- All planned experiments complete (7 Kaggle sessions)
- Project report complete (56 pages, Phase-3 format)
- IEEE conference paper: final draft
- Journal manuscript: complete draft
- Pending: clinician review of candidates

Conclusion and Future Work

Aim 2: accuracy

Beats every baseline on unseen diseases: 76% vs 57% approved drug in top 10.

Aim 1: transfer

Autoimmune data helps only when data is scarce; at full scale it even lowers MRR.

Aim 3: components

Paths carry accuracy; transfer channel adds stability; 72% of edges redundant.

Future work

- Clinician review of top candidates with the prepared review sheet
- KG-embedding baselines and BioPathNet's official code on the same split
- TxGNN comparison on more disease areas; reduce the corticosteroid bias

IEEE Draft and Target Venues

Status of the drafts

- Conference paper (IEEE, 5 pages): final draft
- Journal manuscript (IEEE, about 11 pages): complete draft
- College Turnitin check: 0% similarity; AI score below the reporting threshold
- Every number generated from run logs

Title: BioPathNet++ for Autoimmune Diseases: Path-Based Transfer Learning for Rare Disease Drug Repurposing

Targeted venues (CFPs to be confirmed)

- Conferences: IEEE CONECCT, IEEE INDICON, ICCCNT, InCoB 2027
- Journals: Computers in Biology and Medicine, Journal of Biomedical Informatics, BMC Bioinformatics, IEEE Access, Scientific Reports

References

[1] E. Y. Hu et al., "Enhancing link prediction in biomedical knowledge graphs with BioPathNet," Nature Biomedical Engineering, vol. 10, pp. 1747-1769, 2026.

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[7] D. S. Himmelstein et al., "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," eLife, vol. 6, p. e26726, 2017.

[8] S. Pushpakom et al., "Drug repurposing: progress, challenges and recommendations," Nature Reviews Drug Discovery, vol. 18, pp. 41-58, 2019.

Thank
You